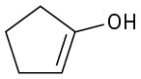
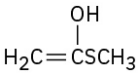
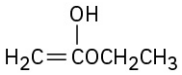
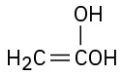
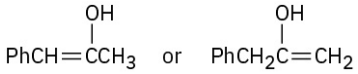
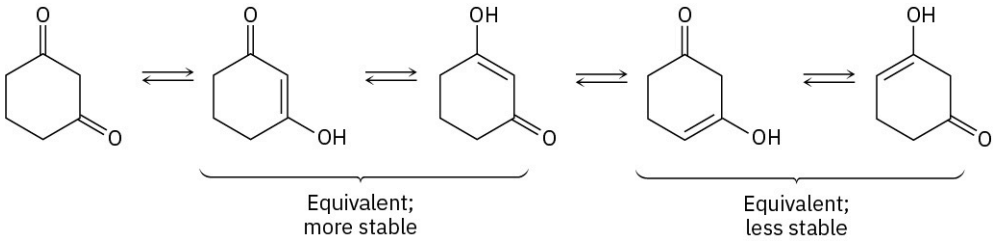
Chapter 22

PROBLEM 22-1 **(a)**  **(b)**  **(c)**  **(d)** $CH_3CH{=}CHOH$ **(e)** 

(f) 

PROBLEM 22-2 **(a)** 4 **(b)** 3 **(c)** 3 **(d)** 2 **(e)** 4 **(f)** 5

PROBLEM 22-3 

PROBLEM 22-4 Acid-catalyzed formation of an enol is followed by deuteronation of the enol double bond and dedeuteronation of oxygen.

PROBLEM 22-5 1. Br_2; 2. Pyridine, heat

PROBLEM 22-6 The intermediate α-bromo acid bromide undergoes a nucleophilic acyl substitution reaction with methanol to give an α-bromo ester.

PROBLEM 22-7 **(a)** CH_3CH_2CHO **(b)** $(CH_3)_3CCOCH_3$ **(c)** CH_3CO_2H **(d)** $PhCONH_2$ **(e)** $CH_3CH_2CH_2CN$ **(f)** $CH_3CON(CH_3)_2$

PROBLEM 22-8 $^{-}{:}CH_2C{\equiv}N{:} \longleftrightarrow H_2C{=}C{=}\ddot{N}{:}^{-}$

PROBLEM 22-9 Acid is regenerated, but base is used stoichiometrically.

PROBLEM 22-10 **(a)** 1. $Na^+\ ^-OEt$; 2. $PhCH_2Br$; 3. H_3O^+
(b) 1. $Na^+\ ^-OEt$; 2. $CH_3CH_2CH_2Br$; 3. $Na^+\ ^-OEt$; 4. CH_3Br; 5. H_3O^+
(c) 1. $Na^+\ ^-OEt$; 2. $(CH_3)_2CHCH_2Br$; 3. H_3O^+

PROBLEM 22-11 Malonic ester has only two acidic hydrogens to be replaced.

PROBLEM 22-12 1. $Na^+\ ^-OEt$; 2. $(CH_3)_2CHCH_2Br$; 3. $Na^+\ ^-OEt$; 4. CH_3Br; 5. H_3O^+

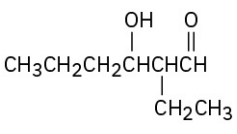
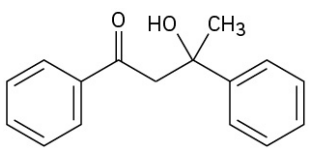
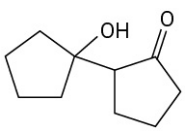
PROBLEM 22-13 **(a)** $(CH_3)_2CHCH_2Br$ **(b)** $PhCH_2CH_2Br$

PROBLEM 22-14 None can be prepared.

PROBLEM 22-15 1. 2 $Na^+\ ^-OEt$; 2. $BrCH_2CH_2CH_2CH_2Br$; 3. H_3O^+

PROBLEM 22-16 **(a)** Alkylate phenylacetone with CH_3I **(b)** Alkylate pentanenitrile with CH_3CH_2I
(c) Alkylate cyclohexanone with $H_2C{=}CHCH_2Br$ **(d)** Alkylate cyclohexanone with excess CH_3I
(e) Alkylate $C_6H_5COCH_2CH_3$ with CH_3I **(f)** Alkylate methyl 3-methylbutanoate with CH_3CH_2I

Chapter 23

PROBLEM 23-1 **(a)**  **(b)**  **(c)** 

PROBLEM The reverse reaction is the exact opposite of the forward reaction shown in Figure 23.2.